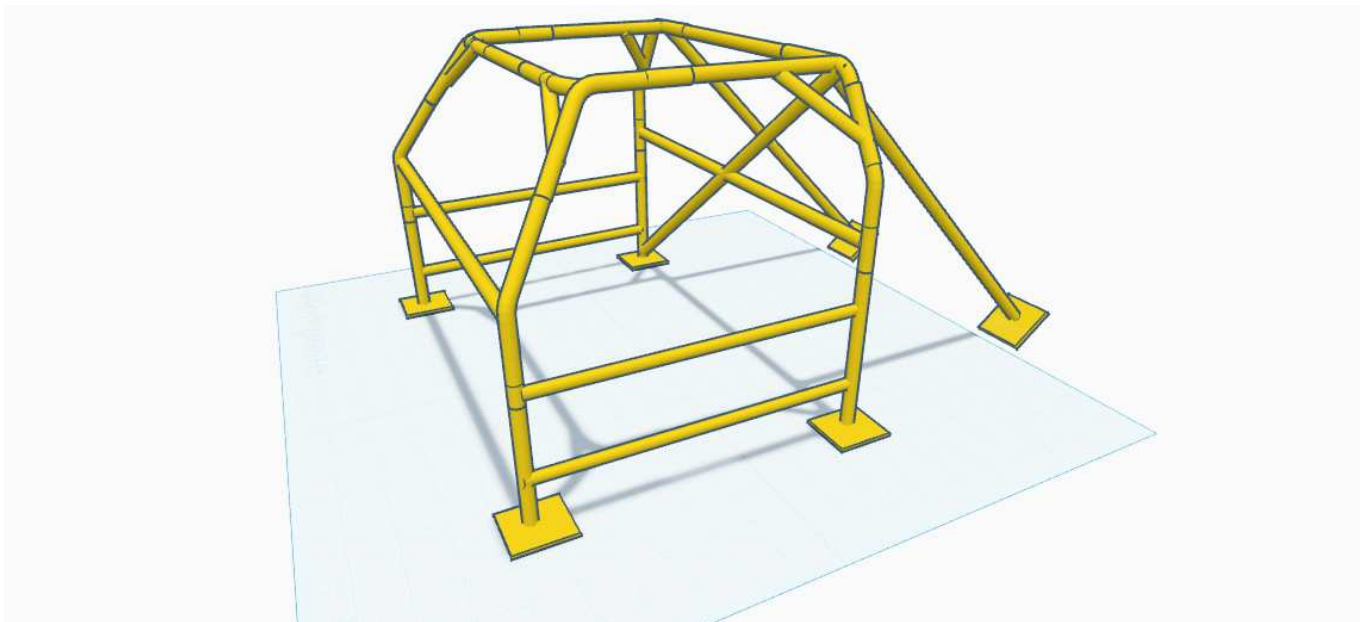


BAVARIAN ENDURANCE BATTLE

Roll Cage Regulations

Construction Inspection and Occupant Protection Requirements



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These regulations establish the minimum roll-cage requirements for vehicles entered in Bavarian Endurance Battle competition. Compliance with the written rules does not replace the responsibility to design, fabricate, install, and maintain a professional competition safety structure.

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Reading the figures. Yellow members represent the base cage structure. A member highlighted in red or green is the component addressed by the adjacent section. A colored member is not automatically optional; the caption and governing text determine whether it is required or supplemental.

Controlling language. Figures explain geometry but do not replace the written requirements. If an illustration and the written rule appear inconsistent, the written rule controls and the team shall request clarification before fabrication.

RC-1 - Purpose Scope and Authority

RC-1.1 Purpose

These regulations establish minimum design, material, fabrication, installation, inspection, and occupant-protection requirements for roll cages used in Bavarian Endurance Battle competition. The objective is a permanently welded safety structure that protects the occupant compartment during rollover, frontal, side, and rear impacts.

RC-1.2 Applicability and Acceptance

Every competing vehicle shall comply with these regulations and pass BEB technical inspection. A cage previously accepted by another motorsport organization may be presented for consideration, but prior approval does not guarantee BEB acceptance. BEB shall conduct its own inspection and may reject any structure that does not satisfy these requirements or presents an unsafe condition.

RC-1.3 Safety Function

The cage shall be designed, constructed, installed, and maintained as an integrated safety system by a professional fabricator, or by a person possessing the knowledge, experience, and equipment required to construct competition roll cages. Compliance shall be evaluated from the completed structure, including:

- Tubing material, outside diameter, and wall thickness
- Main structural members and their placement
- Tube geometry and properly formed bends
- Door and side-impact protection
- Complete 360-degree welded connections at tube junctions and tube-to-plate or tube-to-plinth-box connections
- Mounting plates, plinth boxes, and chassis attachment
- Occupant and helmet clearance
- Approved protective padding
- Overall condition, fitment, workmanship, and craftsmanship

A cage missing a required component, containing noncompliant fabrication, or displaying workmanship inconsistent with a professional motorsport safety structure shall not pass inspection.

RC-1.4 Inspection Authority

BEB may approve, reject, require correction of, or require additional verification for any cage or cage component. Inspection is a condition of participation and is not a certification, warranty, or transfer of design or fabrication responsibility from the entrant, team, owner, driver, or fabricator to BEB.

RC-2 - General Construction and Installation

RC-2.1 Permanently Welded Structure

All roll cages shall be permanently welded structures. Bolt-in, removable, or temporary cages are not accepted. Every required member shall comply regardless of the sequence or method used to install it in the vehicle.

RC-2.2 Professional Fabrication

The cage shall be designed, fabricated, welded, and installed to a professional motorsport-fabrication standard. All tubing shall be accurately measured, properly notched, and fitted before welding. Excessive gaps shall not be filled primarily with welding material.

RC-2.3 Complete Circumferential Welding

Every structural tube junction shall be welded completely around the full circumference of the connection, including tube-to-tube junctions, tubes terminating against intersecting members, and tubes terminating at mounting plates or plinth boxes.

RC-2.4 Chromoly Welding

A cage constructed from seamless 4130 alloy-steel tubing, commonly known as chromoly, shall be welded only by the gas tungsten arc welding process (GTAW or TIG). Adding filler rod is mandatory; fusion of the tube surfaces without filler is prohibited. Accepted filler rods are ER80S-D2, ER70S-2, and ER70S-6.

RC-2.5 Condition and Rejection

The following conditions shall result in rejection:

- Cracked, split, crushed, or severely corroded tubing
- Crimped, kinked, collapsed, or materially flattened bends
- Incomplete, cracked, contaminated, porous, or visibly defective welds
- Poorly fitted tube junctions or excessive gaps
- Damaged or distorted structural members
- Improper repairs or alterations
- Drilled, cut, or removed required members
- Mounting points attached to weakened, torn, or corroded body structure
- Sharp projections capable of contacting an occupant

Any collision, repair, alteration, or damage involving the cage or its mounting points shall be disclosed to BEB before the vehicle returns to the racing surface.

RC-3 - Minimum Cage Configuration

RC-3.1 Six-Point Minimum

The minimum permitted roll cage shall be a permanently welded six-point structure. The six required mounting points consist of two main-hoop mounting points, two forward mounting points, and two rear-brace mounting points. Each required cage member shall terminate at an approved mounting plate or plinth box attached to structurally sound vehicle construction.

BEB strongly recommends adding left and right front-wheel anti-intrusion assemblies. A compliant six-point cage with these two additional mounting points is identified as a 6+2 cage. The designation and construction are explained in RC-3.6.

RC-3.2 Minimum Required Structure

The minimum six-point cage shall include:

- One main hoop
- One continuous main-hoop diagonal
- One permanently welded harness bar
- Two forward roof-and-A-pillar members, or one compliant front hoop
- One upper windshield crossbar when the forward structure uses separate left and right members
- Two mandatory straight rear braces
- Required driver-side door protection
- Required passenger-side door protection
- Six approved mounting plates or plinth boxes
- Four mandatory tubular corner gussets specified in RC-8.10

Supplemental roof reinforcement is recommended but is not a mandatory member of the basic six-point cage.

RC-3.3 Harness Bar

A permanently welded harness bar is mandatory. Its height and geometry shall provide proper shoulder-harness routing for the installed seat and every driver. Complete requirements appear in RC-8.3.

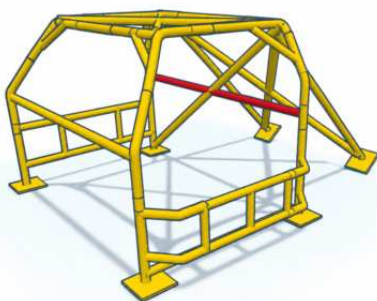


Figure RC-3.3A - Harness-bar location, first perspective.



Figure RC-3.3B - Harness-bar location, alternate perspective.

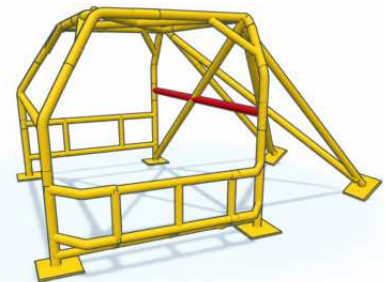


Figure RC-3.3C - Harness-bar relationship to the main hoop.

RC-3.4 Generic Six-Point Cage Configuration

Figures RC-3A through RC-3C show the general layout of the minimum cage. They are orientation drawings and shall be read together with every component requirement in this document.

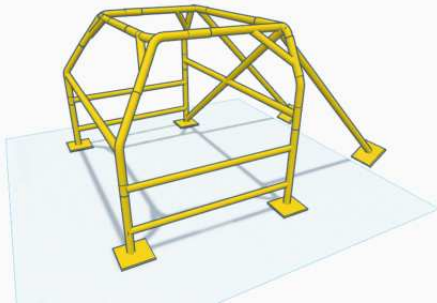


Figure RC-3A - Generic six-point cage, front perspective.

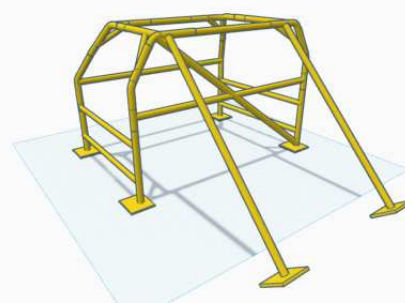


Figure RC-3B - Generic six-point cage, alternate perspective.

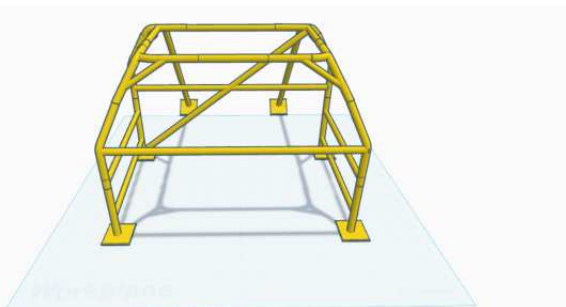


Figure RC-3C - Generic six-point cage, rear perspective.

RC-3.5 Supplemental Members

Additional reinforcement may be installed when it does not reduce occupant clearance, interfere with emergency extraction, create a hazardous projection, or weaken a required member. Supplemental members shall comply with the applicable material, fitment, welding, and workmanship requirements. In supplemental-component figures, red tubing identifies the optional member being discussed.

RC-3.6 Front-Wheel Anti-Intrusion Assemblies and 6+2 Designation

Left and right front-wheel anti-intrusion assemblies are strongly recommended. Each assembly should use two straight tubes extending forward from the straight lower section of the corresponding forward roof-and-A-pillar member. The two tubes shall converge in a V or greater-than-symbol shape and terminate together at a dedicated plate positioned on structurally sound floor or wheel-well structure near the front-tire intrusion area.

Each anti-intrusion plate shall be the same material thickness required for the other cage mounting plates and shall measure at least 4 inches by 4 inches. The assembly is intended to resist inward displacement of the front tire and wheel assembly toward the driver footwell during a collision.

A six-point cage equipped with compliant anti-intrusion assemblies on both front wheel-well areas is a 6+2 cage. Six points remain the required main-hoop, forward, and rear-brace points; the additional two points are the dedicated left and right anti-intrusion plates.

RC-4 - Tubing Materials and Minimum Dimensions

RC-4.1 Accepted Tubing

Required cage members shall be constructed only from one of the following materials:

- Drawn-over-mandrel mechanical steel tubing meeting ASTM A513 Type 5, commonly identified as DOM, in 1020 or 1026 steel or an approved equivalent
- Cold-drawn seamless mechanical steel tubing of equivalent specification
- Seamless 4130 alloy-steel tubing, commonly known as chromoly
- Docol R8 tubing when documentation establishes the material and the tubing meets the applicable minimum dimensions

RC-4.2 Vehicle Weight

The applicable tubing group is determined from the vehicle competition weight declared by the team and accepted or verified by BEB. When the weight is uncertain or falls near a group boundary, BEB may require the larger tubing specification.

RC-4.3 Minimum Tubing by Vehicle Weight and Material

Competition weight	DOM or CDS mild steel	Seamless 4130 chromoly
1,500 lb or less	1.375 in OD x 0.095 in wall or 1.500 in OD x 0.080 in wall	1.375 in OD x 0.080 in wall
1,501-2,500 lb	1.500 in OD x 0.095 in wall	1.500 in OD x 0.080 in wall
2,501-3,000 lb	1.500 in OD x 0.120 in wall or 1.750 in OD x 0.095 in wall	1.500 in OD x 0.095 in wall
3,001-4,000 lb	1.750 in OD x 0.120 in wall	1.750 in OD x 0.095 in wall
More than 4,000 lb	2.000 in OD x 0.120 in wall	2.000 in OD x 0.095 in wall

All required members shall meet or exceed the applicable minimum. Increasing outside diameter or wall thickness does not authorize unsafe geometry, poor fitment, or noncompliant welding.

RC-4.4 Required and Supplemental Tubing

All required cage members shall use the same approved material family and shall meet at least the applicable minimum outside diameter and wall thickness. Supplemental tubing may differ in dimension only when it does not create an unsafe condition and remains suitable for its intended structural function.

RC-5 - Declaration Documentation and Inspection

RC-5.1 Pre-Inspection Roll-Cage Declaration

Before BEB begins visual inspection, the team shall submit a completed roll-cage declaration signed by the team representative. When reasonably available, the fabricating shop or fabricator should also sign the declaration.

The declaration shall identify, at minimum:

- Tubing material
- Tubing outside diameter
- Tubing wall thickness
- Welding process
- Fabricator or shop information when available
- Known collision damage, repairs, or alterations

For an older cage whose original builder or records are unavailable, the team representative shall complete and sign the declaration based on the best available information. Submission of the declaration does not replace or limit BEB visual inspection.

RC-5.2 Visual Inspection

After receiving the completed declaration, BEB shall visually inspect the cage for required components, geometry, fitment, mounting, welding, clearance, padding, condition, and workmanship. The team remains responsible for the accuracy of its declaration and the safety and compliance of the structure.

RC-5.3 Rare Verification Requests

In rare cases, BEB reserves the right to request purchase records, material documentation, builder records, ultrasonic wall-thickness measurement, or another reasonable method of verification. A cage made from tubing that cannot be reasonably verified may be rejected. A required member shall not be drilled for inspection without prior written authorization from BEB.

RC-5.4 Acceptance Is Event-Specific

Passing inspection authorizes participation only under the conditions observed. Acceptance does not prevent later rejection if damage, alteration, deterioration, inaccurate information, or another unsafe condition is discovered.

RC-6 - Tube Bending Fitment Welding and Clearance

RC-6.1 Bending

Bends shall be smooth and formed without kinks, wrinkles, cracking, collapse, or excessive flattening. The centerline bend radius should be at least three times the tubing outside diameter unless the bend is demonstrated to retain equivalent strength and cross-sectional integrity.

RC-6.2 Fitment and Notching

Tube ends shall be accurately notched and closely fitted to the receiving member before welding. The joint shall provide sufficient contact for a sound weld around the entire circumference. Welding material shall not be used as the primary method of filling an excessive gap.

RC-6.3 Weld Quality

Welds shall be free from visible contamination and porosity and shall display steady, uniform beads appropriate to the process and materials. There shall be no burn-through holes or torn sheet metal next to or near a mounting-plate or plinth-box attachment weld. Every tube junction and tube-to-plate or tube-to-box junction shall receive the complete circumferential weld required by RC-2.

RC-6.4 Occupant Clearance

The cage shall provide practical clearance from the driver and shall not force the seat or driver into an unsafe position. With the driver seated in the normal driving position, fully restrained, and wearing the required helmet and frontal head restraint, there shall be at least 2 inches between the top of the helmet and any cage member directly above or capable of contacting the helmet.

RC-7 - Required and Supplemental Members

RC-7.1 Required Members

A required member is part of the minimum cage configuration or is expressly identified as mandatory. It shall not be omitted, replaced by a removable component, or weakened to accommodate interior trim, equipment, wiring, or another vehicle component.

RC-7.2 Supplemental Members

Supplemental members are optional reinforcements. When installed, they become part of the cage and shall satisfy the same applicable material, fitment, welding, mounting, clearance, and workmanship requirements. An optional member shall not interfere with driver visibility, entry, emergency extraction, seat position, harness routing, or helmet clearance.

RC-7.3 Figure Color Convention

In the following figures, yellow generally represents the surrounding cage. Red or green identifies the member or configuration under discussion. The color is a visual aid only. The section text states whether the highlighted component is required, recommended, or optional.

RC-8 - Cage Component Requirements

RC-8.1 Main Hoop

The main hoop shall be one continuous tube extending from one main-hoop mounting point, upward along one side of the occupant compartment, across the roof area, and downward to the opposite main-hoop mounting point. The hoop shall follow the interior body profile as closely as practical while preserving required occupant clearance.

The main hoop may contain no more than four primary bends. A small central relief bend is permitted when necessary to follow a strongly concave roof line; when viewed from the front or rear, this relief may form a shallow inverted-V profile. The lower uprights may be positioned within plus or minus 10 degrees of vertical. The overall hoop should describe a shape as close as practical to a continuous 180-degree loop from one mounting point to the other.

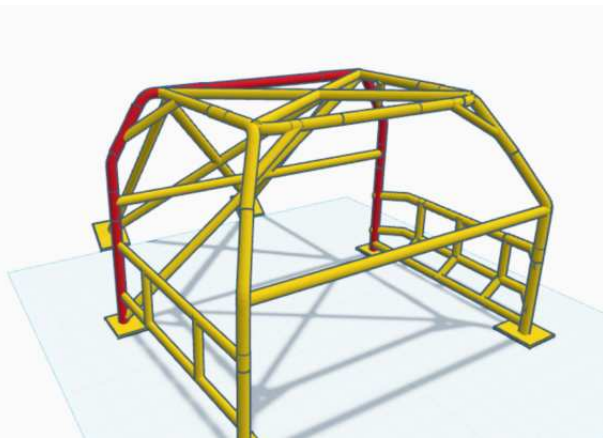


Figure RC-8.1A - Main hoop highlighted, first perspective.

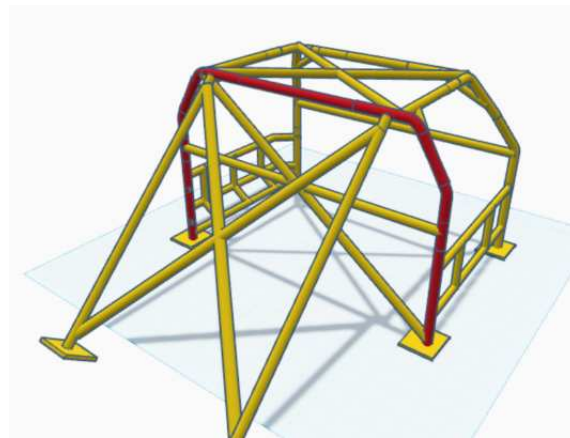


Figure RC-8.1B - Main hoop highlighted, alternate perspective.

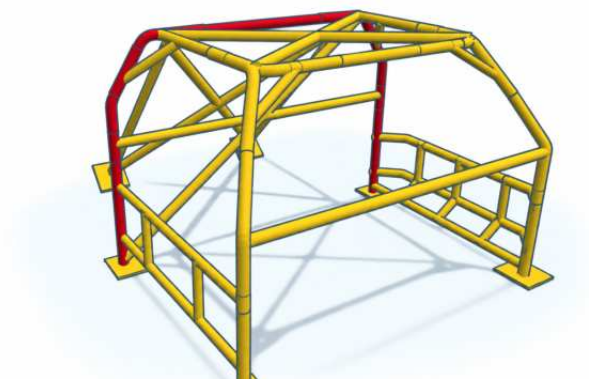


Figure RC-8.1C - Main-hoop path and roof profile.

RC-8.1.1 Driver Helmet Clearance

The two-inch minimum helmet clearance in RC-6.4 applies to the main hoop and every upper cage member. Clearance shall be evaluated with the driver in full competition equipment and normal driving posture.

HELMET CLEARANCE

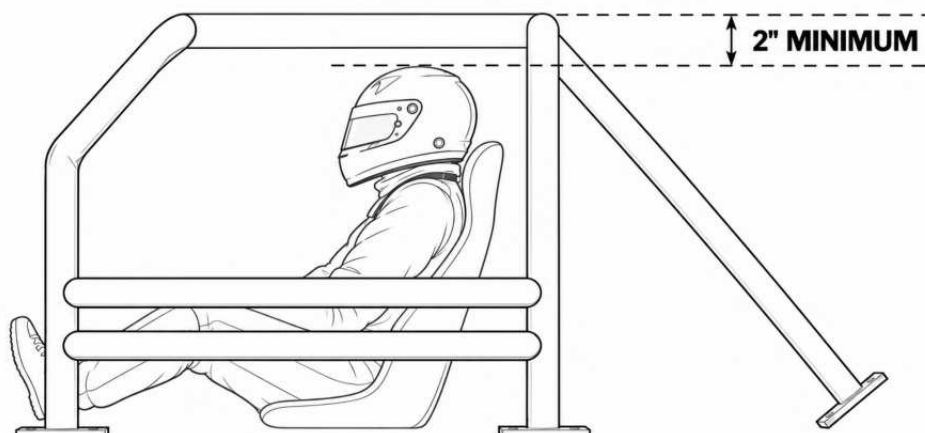


Figure RC-8.1.1A - Minimum two-inch helmet-to-cage clearance.

RC-8.2 Main-Hoop Diagonal

One continuous diagonal member is mandatory within the main hoop. The upper end shall attach at or near the upper main-hoop corner behind the driver and no farther than 12 inches inward from that corner, measured from the hoop bend away from the driver head area.

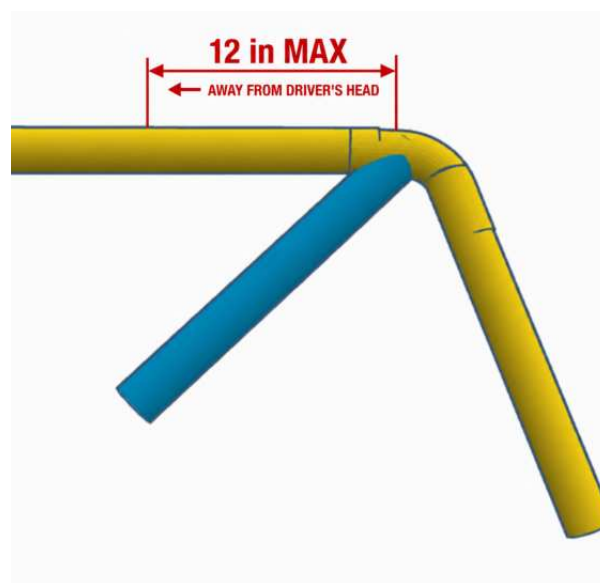


Figure RC-8.2A - Maximum permitted upper attachment distance. Measure no more than 12 inches inward from the main-hoop bend, away from the driver's head.

The opposite end shall attach to the main-hoop mounting plate or to the main hoop as close to that plate as practical at the diagonally opposed corner on the passenger-floor side. The diagonal shall remain one continuous, uninterrupted tube. The harness bar may be divided where it meets the diagonal; the diagonal may not be divided by the harness bar.

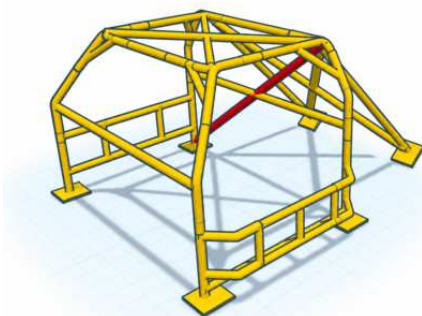


Figure RC-8.2B - Required main-hoop diagonal highlighted in red, perspective view.

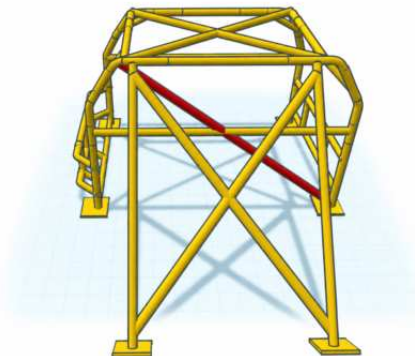


Figure RC-8.2C - Diagonal path between the upper driver-side corner and diagonally opposed passenger-side lower mounting point.

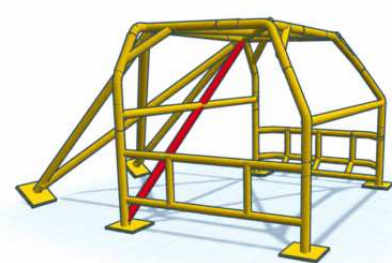


Figure RC-8.2D - Lower diagonal termination at the passenger-side main-hoop mounting plate, or as close to the plate as practical.

RC-8.3 Harness Bar

The harness bar shall be permanently welded between the main-hoop uprights. It may be interrupted only where it intersects the required main-hoop diagonal. In that configuration, the diagonal shall remain one continuous tube and each harness-bar section shall be accurately fitted, aligned, and completely welded between the main hoop and diagonal.

The harness bar may contain up to two bends that offset it rearward to provide seat clearance. The offset shall not change the required bar height or create improper shoulder-harness routing. A bolt-in, removable, or temporary harness bar shall not replace the required welded member.

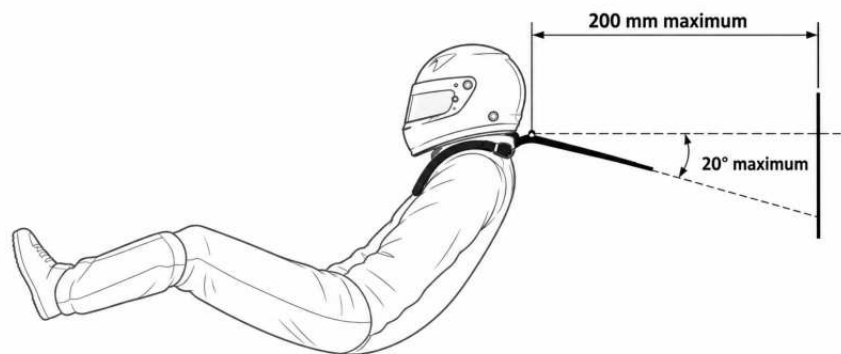
RC-8.3.1 Harness-Bar Height and Shoulder-Belt Angle

The shoulder belts shall travel rearward from the driver shoulders or frontal head restraint to the upper contact surface of the harness bar at an angle from 0 degrees horizontal to 20 degrees downward. The belts shall not angle upward from the shoulders and shall not exceed 20 degrees downward.

Angle and height shall be checked with each endurance driver seated in the normal driving position while wearing the full driving suit, helmet, and required frontal head restraint. Because driver heights differ, the fabricator shall select a bar position that keeps every registered driver within the permitted range. Incorrect routing can increase the risk of spinal, neck, or head injury. The restraint manufacturer instructions shall always be followed.

SHOULDER HARNESS ANGLE

OPTION 1 · TECHNICAL LINE DRAWING



Follow restraint manufacturer's installation requirements.

Figure RC-8.3.1A - Permitted shoulder-belt angle from the driver to the harness bar.

RC-8.3.2 Harness Wrap and Buckle Position

When a shoulder harness is wrapped around the harness bar, the three-bar buckle or adjuster shall be installed in the belt span between the seat and the harness bar, as shown in Figure RC-8.3.2A. The webbing shall be routed through the buckle in the correct sequence so that the completed wrap locks securely under load.

A minimum 100 mm (4-inch) webbing tail shall extend beyond the final pass through the buckle. The completed installation shall remain properly aligned, shall not permit the webbing to loosen or slip, and shall comply with the harness manufacturer's instructions. Twisted webbing or an incorrectly threaded buckle shall not pass inspection.

HARNESS WRAP INSTALLATION

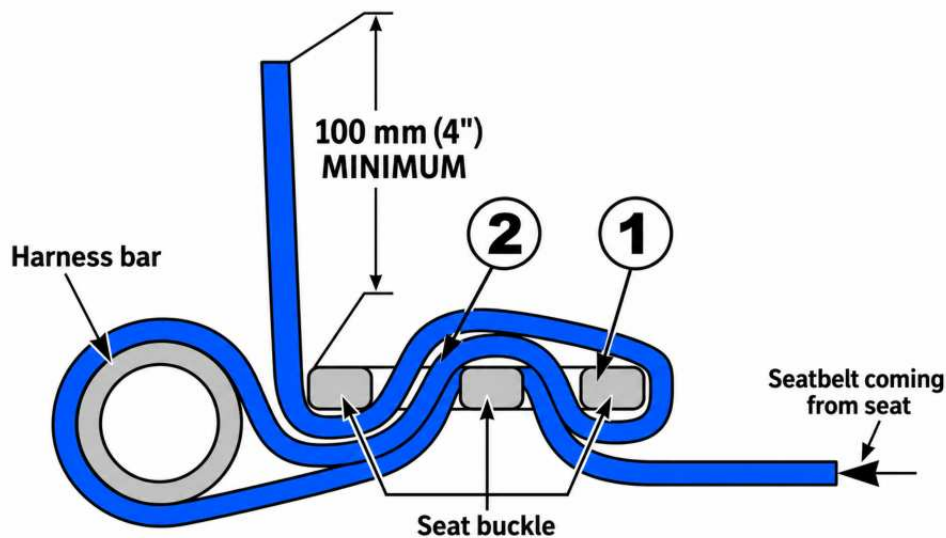


Figure RC-8.3.2A - Correct shoulder-harness wrap and three-bar buckle position between the seat and harness bar.

RC-8.4 Forward Roof-and-A-Pillar Members

The forward structure may use either one continuous front hoop or separate left and right forward roof-and-A-pillar members.

Front-hoop configuration. One continuous tube rises from one forward mounting plate, follows the A-pillar to the roof, crosses the vehicle above the windshield, follows the opposite A-pillar, and terminates at the opposite forward mounting plate. The front hoop shall contain no more than four bends.

Two-member configuration. Each side shall use one continuous tube beginning at its forward mounting plate, extending straight upward through the dashboard area, bending to follow the A-pillar and roof line, and connecting to the main hoop. The section between the mounting plate and the A-pillar bend shall remain straight. The tube may pass through the dashboard when necessary; it shall not be bent around the dashboard.

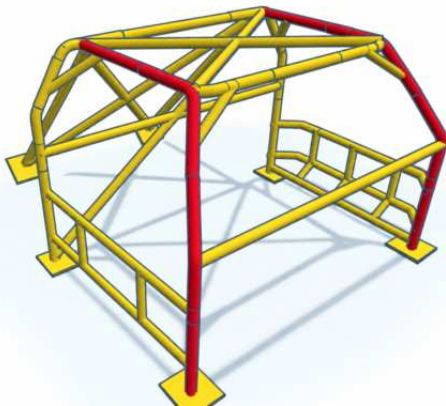


Figure RC-8.4A - Forward roof-and-A-pillar members, first perspective.

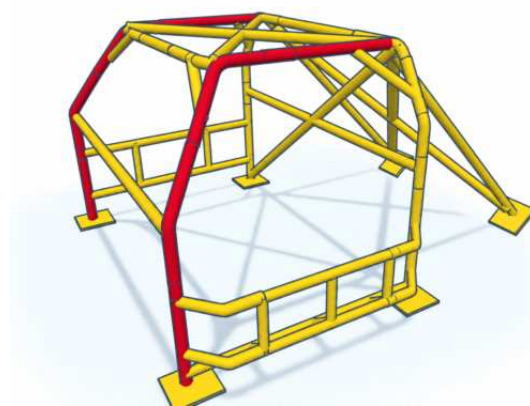


Figure RC-8.4B - Continuous path from mounting plate to main hoop.

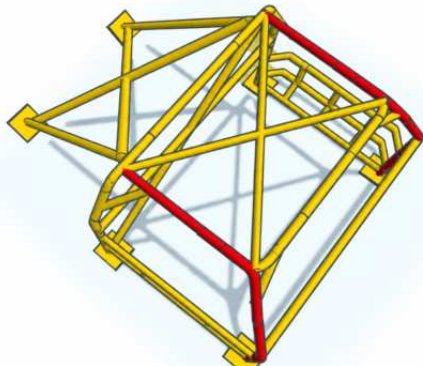


Figure RC-8.4C - Forward-member path, alternate perspective.

RC-8.5 Upper Windshield Crossbar

When separate left and right forward roof-and-A-pillar members are used, one upper windshield crossbar shall connect them above the windshield. The crossbar shall be positioned as close as practical to the area where the windshield meets the roof and shall support the two upper forward corners.

The crossbar may contain up to two bends when needed to follow the windshield or roof contour and improve proximity to the upper body structure. The mandatory tubular corner gussets required by RC-8.10 shall triangulate both upper windshield-crossbar connections.

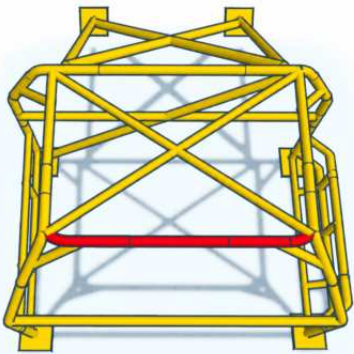


Figure RC-8.5A - Upper windshield crossbar, first perspective.

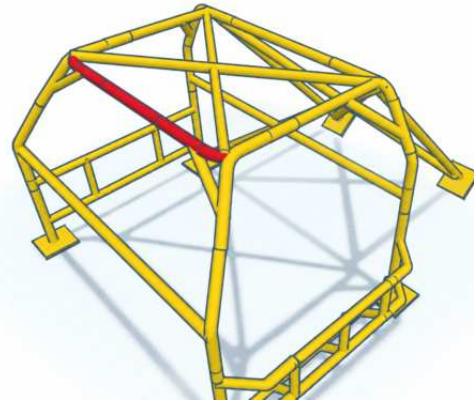


Figure RC-8.5B - Upper windshield crossbar, alternate perspective.

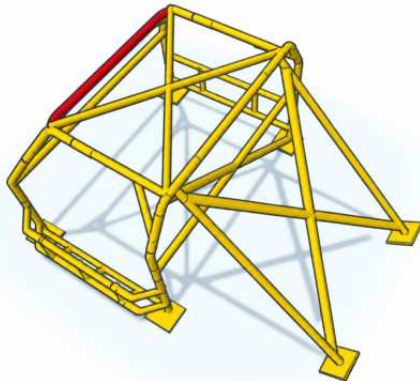


Figure RC-8.5C - Crossbar relationship to the upper forward corners.

RC-8.6 Mandatory Rear Braces Backstays

Two rear braces, also known as backstays, are mandatory. Each brace shall be a straight tube with no bend of any kind. The braces should approach an angle of 45 degrees relative to the plane of the main hoop and should be as nearly parallel to each other as the vehicle structure permits.

Each upper end shall attach as close as practical to the highest portion of the main hoop. When attachment at the top is not practical, the connection shall be no more than 6 inches below the highest point of the hoop.

Each lower end shall terminate on a mounting plate or plinth box attached to a shock tower, frame rail, or another solid floor or chassis point. The plate or box shall be contoured to the supporting surface to obtain the best practical surrounding attachment weld. The rear mounting point shall remain forward of a vertical plane tangent to the rearmost point of the rear-tire tread.

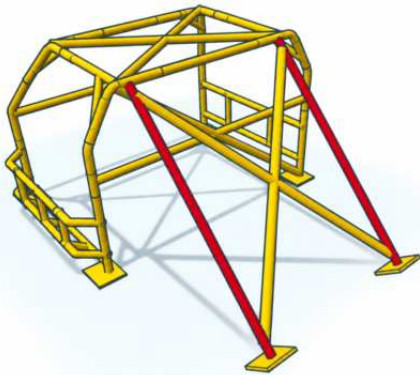


Figure RC-8.6A - Mandatory straight rear braces.

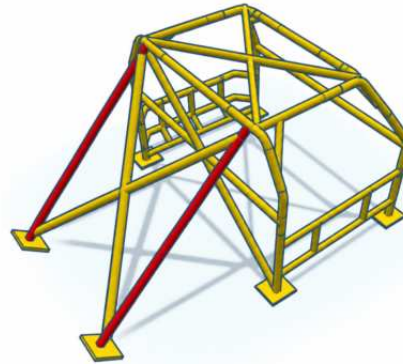


Figure RC-8.6B - Rear-brace angle and upper attachment.

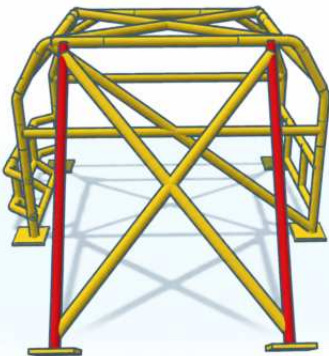


Figure RC-8.6C - Rear mounting-point placement.

RC-8.7 Driver-Side Door Protection

The driver side shall incorporate permanently welded door-bar protection between the main hoop and the forward roof-and-A-pillar member. The system shall extend across the door-opening area, protect the torso, hips, and legs, provide the greatest practical separation from the driver, and permit safe entry and emergency extraction.

Approved configurations are X-type bars, NASCAR-style bars, and two parallel horizontal bars. The driver side may receive greater protection than the passenger side.

RC-8.7.1 Parallel Door-Bar Configuration

When two parallel door bars are used, they shall be separated by at least 7.5 inches measured center-to-center. Both bars shall connect the main hoop to the forward roof-and-A-pillar member. One or more vertical intermediate supports between the bars are strongly recommended.

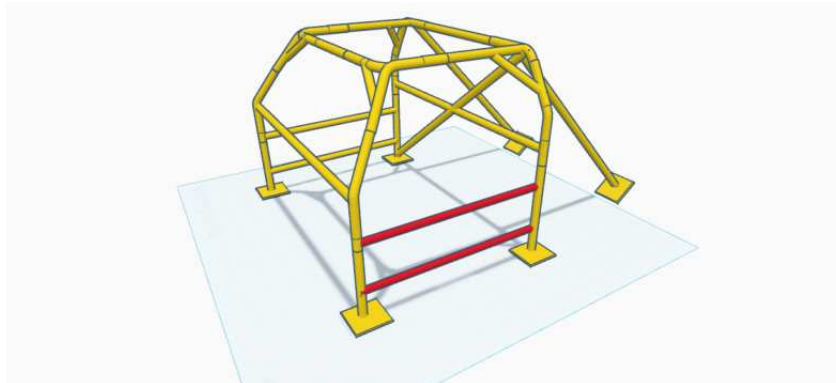


Figure RC-8.7.1A - Parallel door bars highlighted in red.

RC-8.7.2 X-Type Door-Bar Configuration

The X-type configuration shall follow the illustrated geometry. It may be formed as an upper V and lower inverted V whose vertices meet at the center, rather than as two independent tubes simply crossing without a formed central connection. Taco-style gussets shall reinforce the central connection. The center may be shaped outward into the door cavity when this improves driver clearance and egress without weakening the assembly.

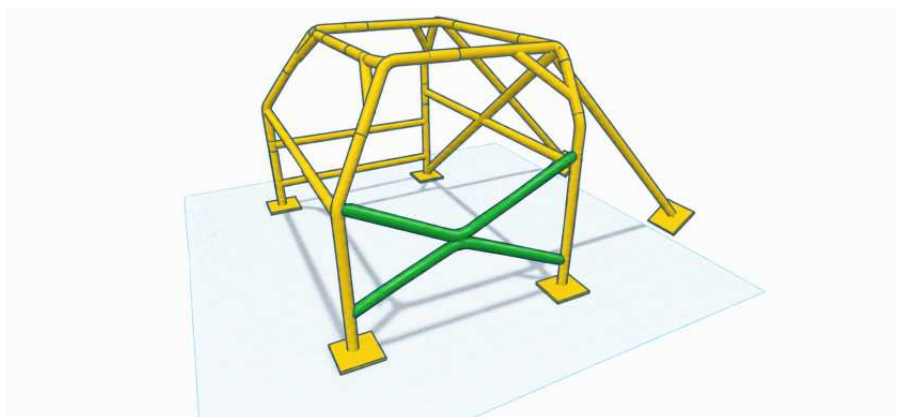


Figure RC-8.7.2A - Approved formed X-type driver-side door-bar configuration. The upper V and lower inverted V meet at the reinforced center connection.

RC-8.7.3 NASCAR-Style Door Bars

NASCAR-style door protection shall use multiple permanently welded bars extending between the main hoop and forward roof-and-A-pillar member, with the primary bars formed outward into the door cavity to increase separation from the driver. Vertical supports shall tie the primary bars together. The complete assembly shall protect the driver torso, hips, and legs without compromising entry or emergency extraction.

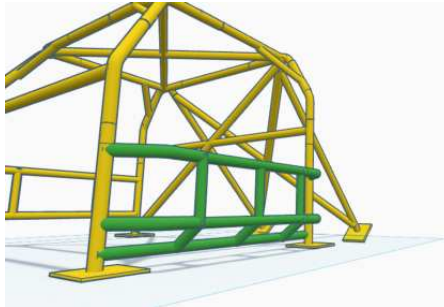


Figure RC-8.7.3A - NASCAR-style door bars, front perspective.

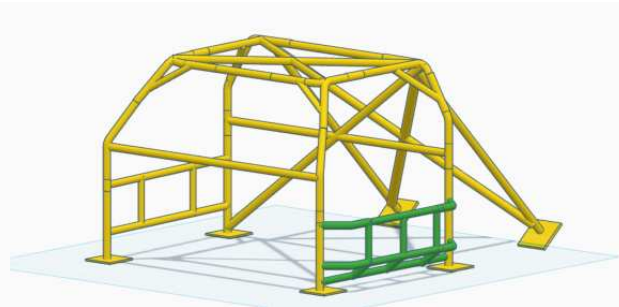


Figure RC-8.7.3B - NASCAR-style door bars, full-cage perspective.

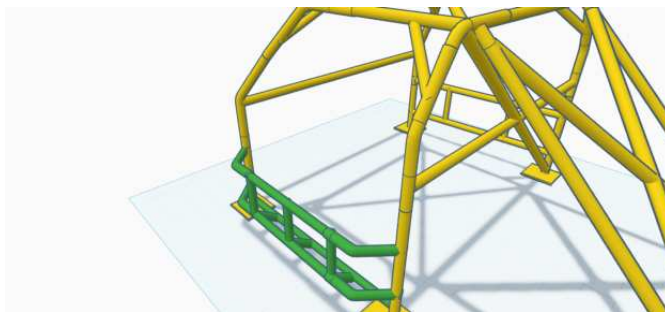


Figure RC-8.7.3C - NASCAR-style bar layout.

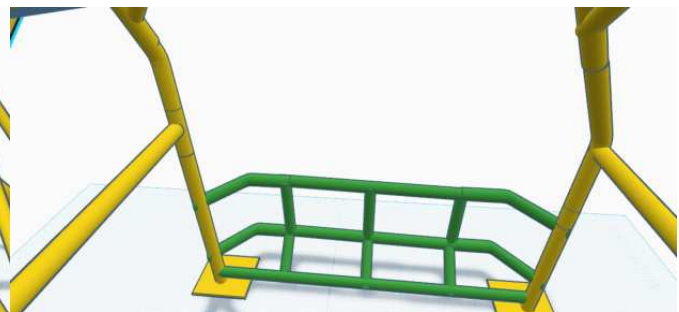


Figure RC-8.7.3D - NASCAR-style bars, interior view.

RC-8.7.4 Lower Sill Bar

A lower sill bar is strongly recommended in addition to the approved driver-side door protection. It should be positioned as low as practical along the rocker or door-sill area and connect the lower portions of the main hoop and forward roof-and-A-pillar member. The sill bar helps resist a low side impact when the striking vehicle nose dives under braking. It is supplemental and shall not replace either required parallel bar or another approved primary door-bar configuration.

RC-8.8 Passenger-Side Door Protection

Passenger-side door protection is mandatory but is not required to be symmetrical with the driver side. The passenger side may use any approved minimum X-type or two-parallel-bar configuration even when the driver side uses NASCAR-style protection. Greater protection on the driver side is strongly recommended because the driver is normally closer to that door.

RC-8.8.1 Asymmetrical Door-Bar Configuration

Figures RC-8.8.1A through RC-8.8.1C illustrate acceptable examples of different configurations on the two sides. Each side shall independently satisfy its applicable minimum requirements.

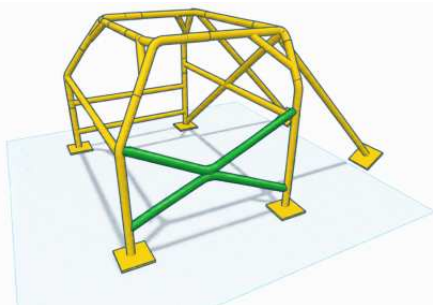


Figure RC-8.8.1A - X-type passenger protection with a different driver-side configuration.

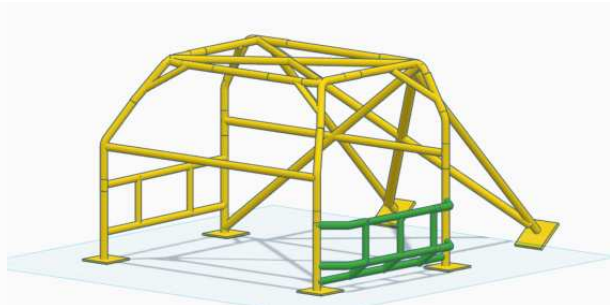


Figure RC-8.8.1B - NASCAR-style driver protection with minimum passenger-side protection.

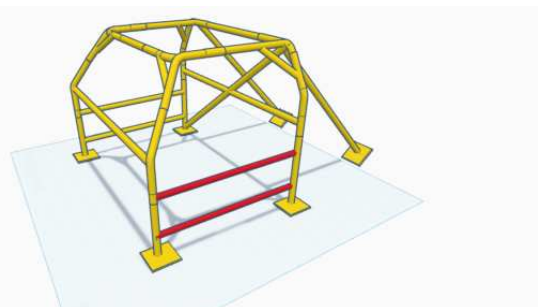


Figure RC-8.8.1C - Parallel-bar passenger-side example.

RC-8.9 Mounting Plates and Plinth Boxes

Every required cage tube shall terminate at an approved mounting plate or plinth box with a complete 360-degree tube-to-plate or tube-to-box weld. A mounting plate or box shall spread impact loads into sound vehicle structure and shall be shaped to conform closely to the supporting surface.

RC-8.9.1 Mounting Locations

The basic six-point cage uses two main-hoop locations, two forward locations, and two rear-brace locations. Each required mounting point shall have its own plate or plinth box. Additional tubes sharing that plate shall attach as close as practical to the required tube.

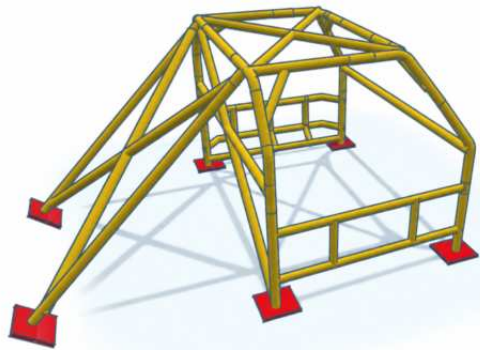


Figure RC-8.9.1A - Required mounting locations, front perspective.

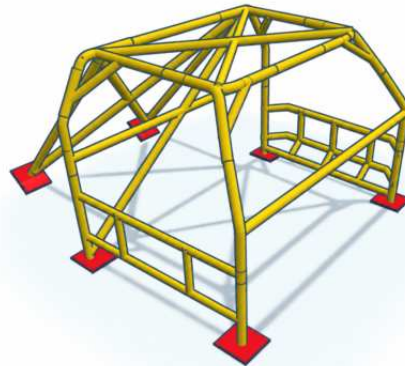


Figure RC-8.9.1B - Required mounting locations, rear perspective.

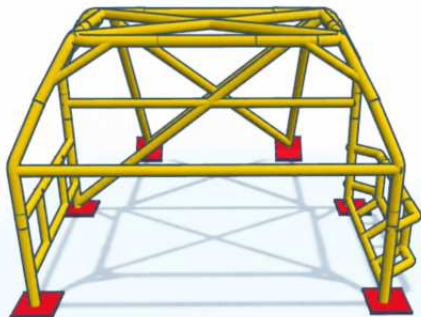


Figure RC-8.9.1C - Required mounting locations, alternate perspective.

RC-8.9.2 Plate Area and Dimensions

Each mounting plate shall have a total area from 9 to 100 square inches. No side shall be shorter than 2 inches or longer than 12 inches. A plate may extend onto vertical structure and may be formed across multiple angles, provided its total developed area does not exceed 100 square inches.

RC-8.9.3 Plate Thickness

A welded steel mounting plate or plinth box shall be at least 0.080 inch thick and no more than 0.250 inch thick. The chosen thickness and shape shall be compatible with the vehicle structure and shall not promote tearing or punch-through.

RC-8.9.4 Attachment to Vehicle Structure

Mounting plates and plinth boxes shall attach to sound body, frame, frame-rail, shock-tower, or floor structure. The attachment welds shall be free from visible contamination and porosity and shall display steady, uniform beads. Thin sheet metal shall not be burned through, torn, or perforated adjacent to the welds. Any hole or tear in the supporting sheet metal near the attachment weld is grounds for rejection.

RC-8.9.5 Plinth-Box Construction

A plinth box shall provide a chassis-contact area equivalent to or greater than an appropriate flat mounting plate and shall remain within the 9-to-100-square-inch total-area range. It shall be contoured to the vehicle structure, fully assembled with sound seams, and attached around its practical perimeter. The required cage tube shall receive a complete 360-degree weld where it terminates on the box.

A bottom support plate is strongly recommended when the box edges could punch through thin sheet metal. The box shall distribute the load into more than a narrow edge, corner, or isolated unsupported panel.

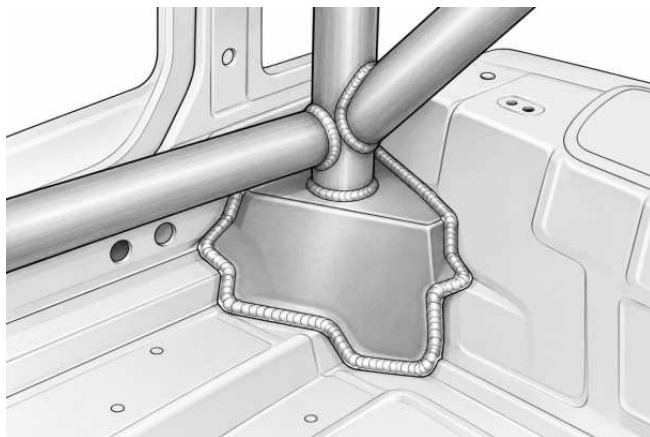


Figure RC-8.9.5A - Contoured plinth box and tube attachment.

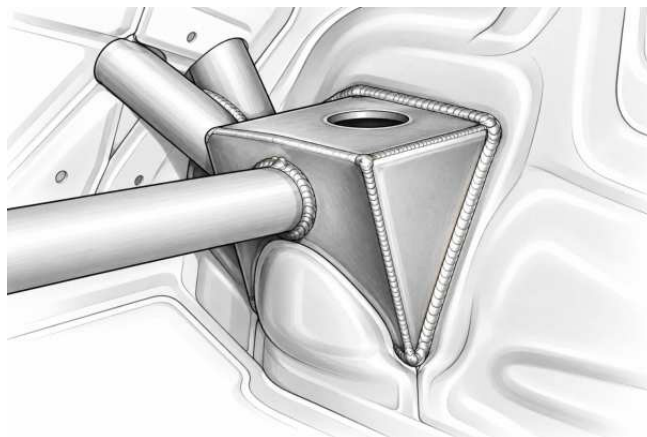


Figure RC-8.9.5B - Plinth-box perimeter and tube weld detail.

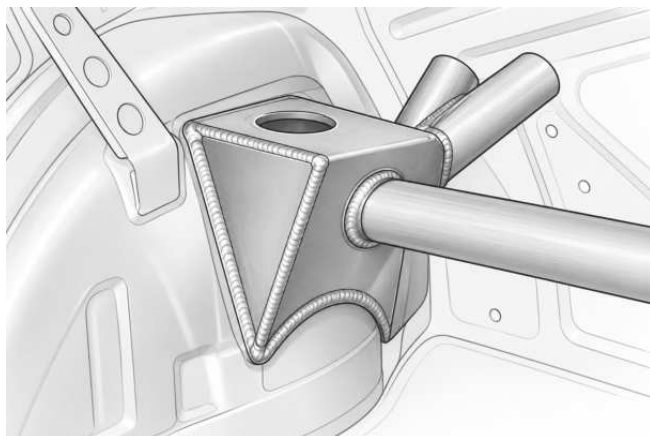


Figure RC-8.9.5C - Multi-surface plinth-box construction.

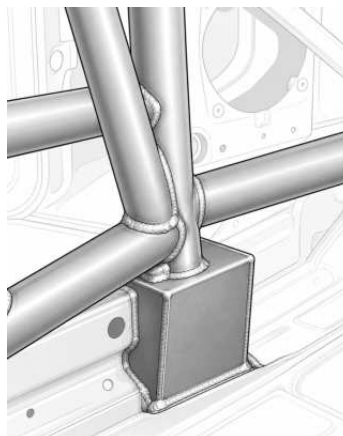


Figure RC-8.9.5D - Plinth box fitted to vehicle structure.

RC-8.10 Mandatory Tubular Corner Gussets

Four tubular corner gussets are mandatory: two at the upper windshield-crossbar connections to the forward roof-and-A-pillar members, and two at the upper connections between the main hoop and the forward roof-side members.

Each gusset shall be a short tube positioned at approximately 45 degrees to triangulate the corner. It shall use the same material as the required cage structure, meet the applicable cage specifications, be accurately notched and fitted, and be welded completely around every notched connection. Plate-style gussets may supplement but shall not replace the four required tubular gussets.

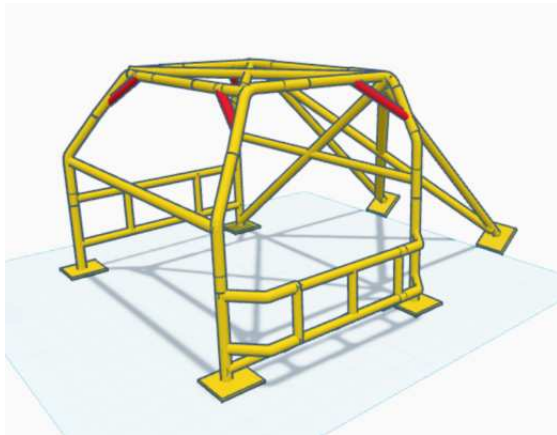


Figure RC-8.10A - Required tubular corner-gusset locations.

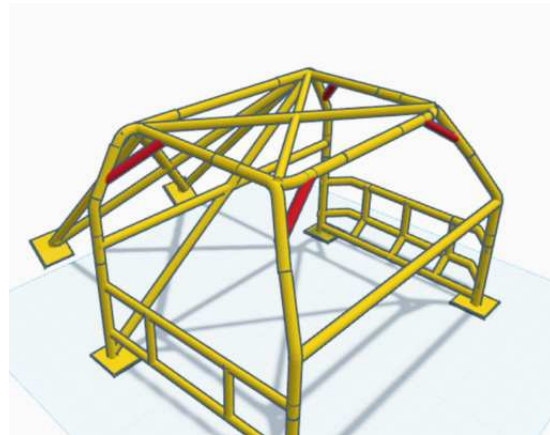


Figure RC-8.10B - Tubular gusset construction and placement.

RC-9 - Supplemental Cage Components

The members in this section are supplemental unless a section expressly states otherwise. If installed, they shall comply with the same applicable material, fitment, notching, welding, mounting, clearance, and workmanship standards as the required cage.

RC-9.1 Dash Bar

A dash bar is highly recommended but not mandatory. It shall be one continuous tube running across the dashboard line and connecting the left and right forward roof-and-A-pillar members at approximately dashboard height. It may contain no more than two bends when an offset is needed for clearance. In Stock class, the bar may be concealed behind the original dashboard. It may pass through dashboard trim when necessary.

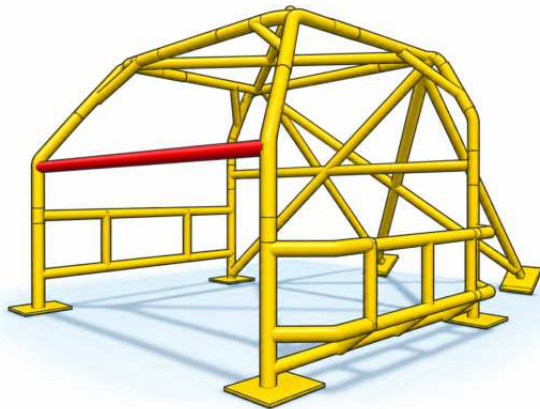


Figure RC-9.1A - Supplemental dash bar, front perspective.

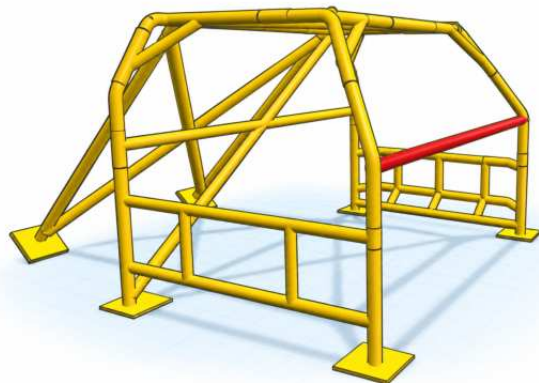


Figure RC-9.1B - Supplemental dash bar, rear perspective.

RC-9.2 Rear-Brace Reinforcement

Supplemental reinforcement may be added between the two mandatory rear braces when vehicle design and occupant clearance permit.

RC-9.2.1 Rear-Brace X

Two supplemental members may form an X between the rear braces, crossing near the center of the area bounded by the main hoop and rear braces. An additional lower crossbar may connect the shock towers or lower rear mounting structure and complete the lower side of the reinforced area. These members do not replace either mandatory straight rear brace.

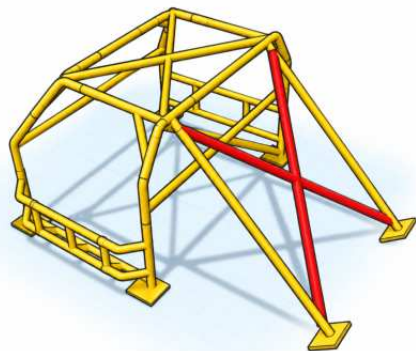


Figure RC-9.2.1A - Supplemental rear-brace X, first perspective.

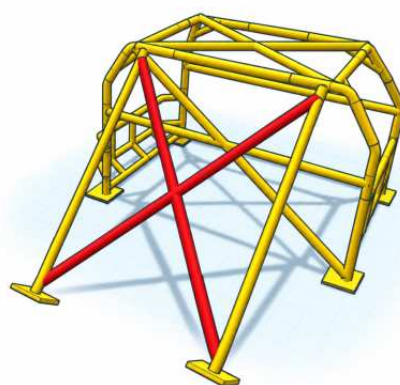


Figure RC-9.2.1B - Supplemental rear-brace X, rear perspective.

RC-9.3 Roof Reinforcement

Roof reinforcement is strongly recommended but is not mandatory when it would reduce required helmet clearance or create interference above the driver. Its design shall preserve the two-inch minimum clearance required by RC-6.4 and RC-8.1.1.

RC-9.3.1 Roof-X Configuration

A supplemental roof X may be installed within the area bounded by the main hoop, forward roof-and-A-pillar members, and upper windshield crossbar. The members may be bent upward toward the roof to increase clearance from the driver helmet, provided the bends are smooth and do not compromise the structure.

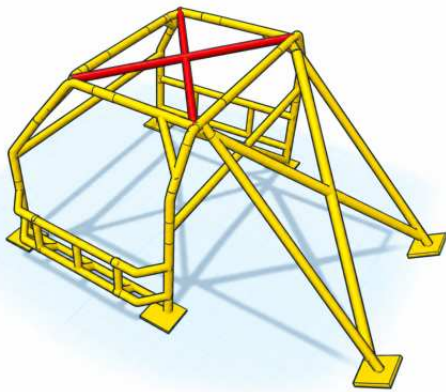


Figure RC-9.3.1A - Supplemental roof-X configuration, perspective view.

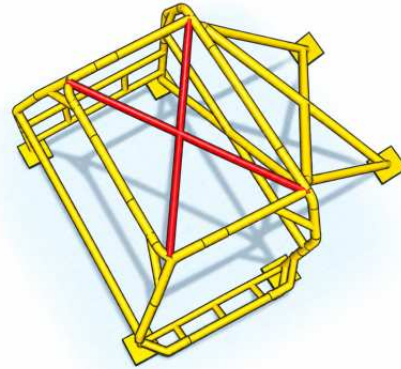


Figure RC-9.3.1B - Supplemental roof-X configuration, overhead view.

RC-9.3.2 Driver Clearance

If either roof-X member crosses too close to the driver helmet, interferes with the two-inch minimum clearance, or creates an unsafe contact path, that member shall be omitted or the roof reinforcement shall use the single-diagonal configuration in RC-9.3.3. Occupant clearance takes priority over installing the optional full X.

RC-9.3.3 Single Roof Diagonal

A single supplemental diagonal may extend from one corner of the roof opening to the diagonally opposite corner. It shall be arranged to avoid the driver head area and preserve the required two-inch helmet clearance. The diagonal shall connect the surrounding roof structure at properly fitted and completely welded junctions.

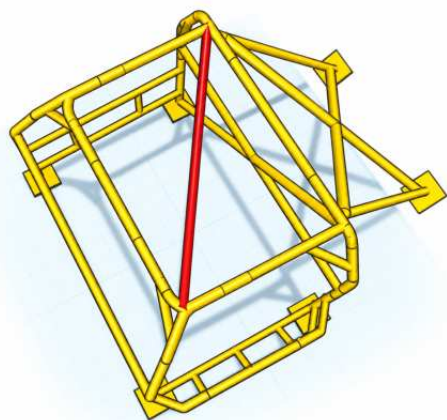


Figure RC-9.3.3A - Supplemental single roof diagonal, overhead perspective.

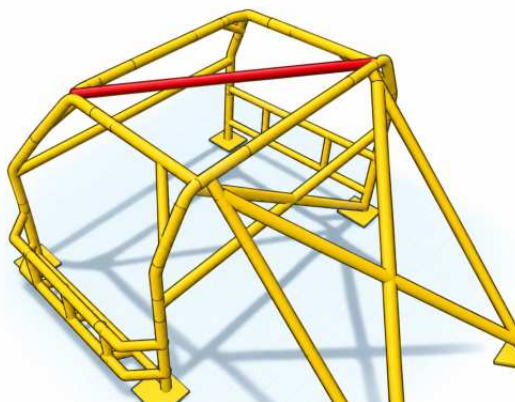


Figure RC-9.3.3B - Supplemental single roof diagonal, alternate perspective.

RC-9.4 Additional Supplemental Reinforcement

Any number of additional reinforcing members may be installed within the cage when they do not create an unsafe condition. A supplemental member shall not replace a required member, reduce required clearance, obstruct emergency extraction, or terminate on unsupported sheet metal.

The examples below highlight supplemental members in red. Their placement is illustrative; the written requirements governing the surrounding mandatory structure remain controlling.

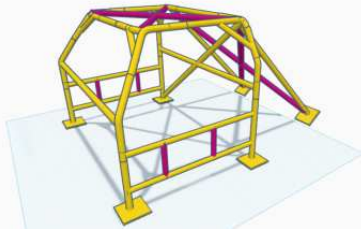


Figure RC-9.4A - Supplemental reinforcement highlighted in red.

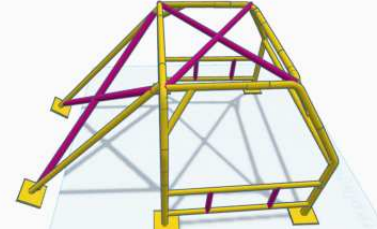


Figure RC-9.4B - Supplemental rear and roof reinforcement.

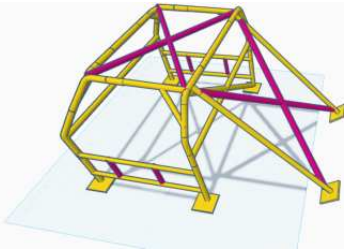


Figure RC-9.4C - Supplemental reinforcement, alternate perspective.

RC-10 - Roll Cage Use and Protective Padding

RC-10.1 Safety Use Only

The roll cage is a dedicated occupant-safety structure. It shall not be used to transport, contain, route, or distribute fuel, oil, coolant, brake fluid, fire-suppression agent, another fluid, vapor, pressurized gas, compressed air, electrical wiring, electrical current, hoses, pipes, control cables, or communication cables.

No cage member may serve as a fluid reservoir, pressure vessel, electrical conduit, wiring channel, or grounding path.

RC-10.2 Drilling and Internal Routing Prohibited

Roll-cage tubing shall not be drilled, pierced, slotted, or otherwise penetrated to route or secure hoses, lines, pipes, wires, or cables. Nothing may be routed through the interior of a cage tube. Required members shall remain sealed and structurally intact.

RC-10.3 Required Protective Padding

Every portion of the cage that the driver helmet, head, arms, legs, or body could contact while the driver is seated and properly restrained in the normal driving position shall be covered with approved high-density impact padding.

Accepted padding shall comply with either FIA Standard 8857-2001 Type A and appear on FIA Technical List No. 23, or comply with SFI Specification 45.1.

Pool noodles, upholstery foam, pipe insulation, soft open-cell foam, cloth or fabric wrapping, carpet, recreational padding, and unrated padding are prohibited.

RC-10.4 Padding Installation

Padding shall match the tube diameter and shape, cover the tube surface facing the driver, remain securely fixed during competition, and be installed so that it cannot rotate, slide, separate, or expose bare tubing. Damaged, loose, deteriorated, or displaced padding shall be replaced.

Manufacturer-approved contact adhesive is recommended between the padding and tube. Multiple heavy-duty cable ties or another approved mechanical fastening method should then secure the padding. Cable-tie heads and other fasteners shall face away from the driver, and trimmed ends shall be cut flush so they do not create sharp projections.

Padding capable of rotating around the tube, moving out of position, or exposing the driver to bare tubing shall not pass inspection.

Publication Checklist

Before presenting a vehicle for BEB roll-cage inspection, the team should verify each of the following:

- ☐ Completed and signed roll-cage declaration
- ☐ Approved tubing material and correct minimum dimensions
- ☐ Permanently welded six-point minimum structure
- ☐ Main hoop, diagonal, harness bar, forward structure, windshield crossbar, and two straight rear braces
- ☐ Required driver-side and passenger-side door protection
- ☐ Six compliant mounting plates or plinth boxes
- ☐ Four mandatory tubular corner gussets
- ☐ Complete circumferential tube-junction and tube-to-plate or tube-to-box welds
- ☐ Correct harness-bar height for every driver
- ☐ Minimum two-inch helmet clearance
- ☐ Approved FIA or SFI padding secured against rotation
- ☐ No drilling, internal routing, fluid transport, or electrical use of cage tubing
- ☐ All collision damage, repairs, and alterations disclosed to BEB

Technical questions. When a chassis condition, unusual body shape, or proposed configuration is not clearly addressed, obtain BEB guidance before fabrication. Approval of a drawing or concept does not excuse noncompliant materials, welding, mounting, clearance, or workmanship.

For recommendations, comments, questions, or concerns regarding these regulations, please email info@bavarianendurancebattle.com.

END OF ROLL CAGE REGULATIONS